

Skinny Physics for QS&BB

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1 Introduction to Skinny Physics

I'm guessing that you might come from various backgrounds and since posts in **QS&BB** may rely on a small bit of background, I've created this Skinny Physics collection to help you whether you're an English major with little physics in your life to an engineer with more.

1.1 Chapter structure

Each chapter has the same structure in support of your particular level of preparation:

1. **Just the facts.** The equations for those of you who might have had some physics in your background and need a refresher
2. **Different way.** There are some places where I introduce ways of approaching things in QSBB that you wouldn't have seen elsewhere. So follow those links for sure.
3. **Gentle explanations of...** This is textbook-like, but textbook-lite. From the beginning but not overpowering. Some videos of explanation.
4. **Pointers** to topics so you can quickly go to a topic where you need more than just the equations. That's #4 just above

The Parts are:

1. Mechanics
 1. Motion
 2. Momentum and Force
 3. Collisions

- 4. Energy
- 2. Gravitation
 - 1. Copernicus' heliocentric proposal
 - 2. Galileo's Astronomy
 - 3. Kepler's Astrophysics
 - 4. Newton's Gravitation
- 3. Electricity and Magnetism
 - 1. Electric Charge and Magnetism
 - 2. Faraday's Experiments and Conclusions
 - 3. Maxwell's Theory
 - 4. Forces on electrical charges
- 4. Einstein's Theory of Special Relativity

Part I

Part I: Mechanics

This Part introduces mechanics — the core of classical physics and a source of important concepts for **QS&BB**, not to mention a lot of vocabulary of physics...velocity, momentum, force, energy, and so on.

Read or skim, depending on your background. So many of these concepts have stood the test of time. As always, my goal is to get you ready for the 21st century physics of particle physics and cosmology.

2 Skinny Motion, Speed, and Acceleration

This is the first of three outlines of some basic physics ideas so that I can refer to them in other posts. If you'd like more, including history and examples, then visit [full textbook](#) for a textbook-level presentation. If you only need the basic simple equations, they're here. If you'd like some explanation, that's next!

Units. Sorry. In QS&BB we won't care about English versus metric units but for this motion review we'll sometimes have to convert. I'll do it for you but you can check me at places like [unitjuggler](#)...or often just Google.

2.1 Different way

Important

Even if you know some of the basics of velocity and acceleration, you need to be comfortable with spacetime diagrams in Section ?? ! That's how we'll pursue Feynman Diagrams.

2.2 Just the facts:

changes in something

I'll use: